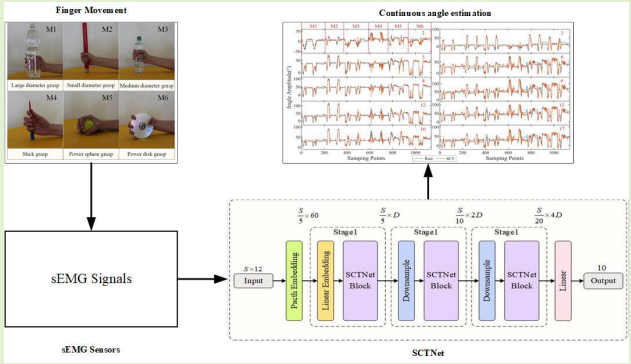
SCTNET: Shifted Windows and Convolution Layers Transformer for Continuous Angle Estimation of Finger Joints Using sEMG

Zida An, Fei Wang, *Member, IEEE*, Chuanjian Ren, Kun Zhang, Yao Pei, Qichuan Ding, *Member, IEEE*, and Shuai Han

***Abstract*—Using noninvasive technology to control prosthetics remains a real-life challenge due to the low acceptability of prosthetics stemming from their unnatural motion trajectory. The continuous angle estimation of fingers based on surface electromyography (sEMG) can effectively improve the unnatural motion trajectory. This article proposes the transformer using shifted windows and convolution networks (SCTNets) to achieve high-precision, fine-grained, and high naturalness finger continuous angle estimation. The shifted windows are used to encrease receptive field and capture multiscale sEMG signals. The transformer is used to capture the long-term dependencies within multiscale sEMG signals, facilitating the extraction of global features, while CNN is used to extract local features. We use 20 subjects of the Ninapro DB2 dataset to test the model, and compared with LS-TCN, CNN-Attention, ConvLSTM and long short-term memory network (LSTM) models. The Pearson correlation coefficient (PCC), the normalized root-mean-square error (NRMSE), and the coefficient of determination (R^2) of the SCTNet was 84.35%, 8.85%, and 69.65%. Compared with the indicator results of other models, the SCTNet has improved by 1% (PCC), 0.15% (NRMSE), and 2.13% (R^2) in each index, and achieved competitive results in real-time computing performance. The results indicate that the SCTNet can effectively estimate the natural continuous finger joint angles.**



***Index Terms*—Continuous angle estimation, finger joint, prosthetics, shifted windows and convolution network (SCTNet), surface electromyography (sEMG).**

I. Introduction

At present, robotic prosthetics have greatly affected the quality of life of people with disabilities [1], [2]. However, numerous challenges still lie ahead. In most cases, the executable actions of a prosthetic hand are limited to opening and closing. In recent years, there has been a surge in the development of humanoid-designed mechanical prostheses capable of executing various programmed movements. However, the control methods for these prostheses are relatively rudimentary, relying on sequential control strategies. This unnatural movement necessitates significant user training and familiarization, imposing substantial usage costs. Consequently, this factor contributes significantly to the current low acceptance of mechanical prosthetics. Due to the rapid development of wearable devices, the user's movement intention can be obtained through noninvasive devices. However, one of the current challenges is how to use wearable sensors for efficient and accurate motion intention recognition.

Surface electromyography (sEMG) signal, as a physiological signal closely related to human movement, is one of the means to achieve efficient and accurate movement intention recognition [3]. sEMG signal comes from human body, and its acquisition is simple and mature [4]. sEMG signal is generated ahead of the motion, which has certain prediction advantages and will provide greater safety margin than the

Manuscript received 10 May 2024; revised 2 July 2024; accepted 2 July 2024. Date of publication 15 July 2024; date of current version 15 August 2024. This work was supported in part by the Foundation of National Natural Science Foundation of China under Grant 62373086 and Grant 62373087. The associate editor coordinating the review of this article and approving it for publication was Dr. Theerawit Wilaiprasitporn. *(Corresponding authors: Fei Wang; Shuai Han.)*

Zida An, Fei Wang, Chuanjian Ren, Kun Zhang, and Qichuan Ding are with the Faculty of Robot Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: 2110702@stu.neu.edu.cn; wangfei@mail.neu.edu.cn; 2070752@stu.neu.edu.cn; 2302186@stu.neu.edu.cn; dingqichuan@mail.neu.edu.cn).

Yao Pei is with the College of Information Science and Engineering, Northeastern University, Shenyang 110819, China (e-mail: 2370774@stu.neu.edu.cn).

Shuai Han is with the Department of Neurosurgery, Shengjing Hospital of China Medical University, Shenyang 110055, China (e-mail: hanshuai19870217@outlook.com).

Digital Object Identifier 10.1109/JSEN.2024.3423795

1558-1748 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

physical signal. The sEMG signal is controlled by cerebral cortex, which is suitable for motor intention estimation of patients with physical disability. Morita et al. [5] proposed a direct torque control method for prosthetic hand. An artificial neural network based on feedback error learning mode is used to estimate joint torque from EMG signals. Raj et al. [6] proposed a proportional–integral–derivative (PID) control method driven by sEMG, which can control the prosthetic hand model in real-time according to human intention. The proposed method is simple in design and can be implemented on people at low cost. Research on recognition of human motion intention based on sEMG signals is mainly divided into discrete motion classification and continuous motion estimation [7].

Discrete motion classification is to recognize the discrete motion modes of prosthetic hand gestures based on sEMG signals. Pomboza-Junez and Terriza [8] proposed support vector machine (SVM) as a classification method to realize gesture recognition. Gaussian radial basis (RBF) is the best kernel function, and the classification accuracy is 79.36%. Park and Lee [9] proposed a user adaptive decoding method, which uses CNN for deep feature learning. The method based on CNN shows better performance than SVM in terms of nonadaptation and adaptation. In particular, the adaptive performance of the proposed method based on CNN shows a classification accuracy of more than 90%. Rahimian et al. [10] proposed a framework based on transformer architecture to process sEMG signals, classify, and recognize upper limb gestures and achieved 82.93% accuracy in the Ninapro DB2 dataset. Lv et al. [11] proposed a multiattention multiperspective (MMDL) multiattention deep learning framework for gesture recognition using sEMG. The average intratarget gesture recognition based on the myo dataset, myoup dataset, and Ninapro DB5 increased by 0.46%, 18.88%, and 7%, respectively. However, discrete motion classification belongs to coarse-grained motion classification, which will lose the motion information in the motion mode, lack of natural motion angle, and increase the difficulty of users.

Continuous motion estimation based on sEMG signals can be divided into methods based on biomechanical models and regression models [12]. Xi et al. [13] used a state-space model combined with the forward dynamics and Hill-based muscle model (HMM) to estimate the continuous motion of the knee–ankle joint of the lower extremity. However, because the method for building the human muscle model is complex, it is challenging to measure the relevant physiological parameters because there are so many of them, and because the physiological characteristics of muscles will change significantly in actual exercise [14], related research has focused more on techniques based on regression models. With the development of deep learning, more and more researchers have begun to apply deep learning to the research of continuous motion estimation based on sEMG signals. Alazrai et al. [15] proposed a method based on support vector regression (SVR) for continuous estimation of wrist and finger joint angles from sEMG signals. The experimental results show that the proposed method can achieve an average regression accuracy (R^2) of 0.64. Ma et al. [16] proposed four widely used deep learning models, including alexnet, RESNET, LSTM, and Gru, for continuous finger motion estimation based on sEMG. Alexnet can make better use of sEMG function (rms) and is superior to the other three networks. (correlation coefficient = 0.7330 ± 0.0572). Wang et al. [17] and Guo et al. [18], respectively, proposed a long short-term memory network (LSTM) and a long exposure convolutional memory network (LE-ConvMN) to predict the angles of ten joints of the hand.

Although LSTM has the ability to retain historical data, it must operate serially based on sequence data, which results in a high time complexity of the model and makes it unsuitable for handling long-term dependencies [19]. Chen et al. [20] proposed the LS-TCN model and used it in the Ninapro DB2 dataset for continuous motion estimation of ten joint angles of the fingers. The TCN model extracts time-series information by dilation causal convolution and residual connection, and experiments prove that it has stronger modeling ability than the LSTM model. Vaswani et al. [21] proposed that the transformer model establishes the global dependencies of sequence data through the self-attention mechanism and uses absolute position encoding to introduce sequence information to the model, which has achieved good results in the field of natural language processing (NLP). Transformer was first used in computer vision by Dosovitskiy et al. [22], and the proposed ViT model produced excellent results in several typical vision tasks. However, because the computational complexity of the transformer model is quadratically correlated with the length of the sequence, Liu et al. [23] proposed that the Swin Transformer model effectively reduces computational complexity through sliding window self-attention and was once in the leading position in visual field tasks such as image segmentation and target detection. We attempt to introduce Swin Transformer into the field of physiological signal analysis and conduct an in-depth study on its improved method in sEMG signal-based continuous motion estimation.

In this article, we propose an shifted windows and convolution network (SCTNet)-based continuous motion estimation framework to estimate finger joint angles. We use shifted window based self-attention (SW-MSA) to model the global dependencies of sequence data, use layer-by-layer downsampling to improve the receptive field range of features, and learn the timing information of sEMG signals through learnable relative position encoding. The we propose a locally enhanced feedforward neural network (LEFFN) layer, which combines the advantages of CNN for extracting local features and transformer for extracting global features and enhances the representation ability of features to improve the accuracy of motion estimation. The model is verified on the currently widely used Ninapro DB2 dataset, and other methods [17], [18], [20], [24] on the dataset are reproduced for comparison. Finally, ablation experiments are designed to illustrate the importance of each part of the model.

The innovation of this article includes the following content.

1) We propose a framework for continuous estimation of finger joint angles based on SCTNet. This framework has leveraged a shifted window data augmentation method to enhance sEMG features and used a self-attention mechanism to better use these features for estimation purposes.

2) The proposed method has been tested on 20 subjects from the Ninapro DB2 dataset. Comparative analysis against LS-TCN, CNN-Attention, ConvLSTM, and LSTM models has demonstrated its superior performance. In comparison to the current state-of-the-art results, our method has achieved the highest improvements of 7.92%, 4.19%, and 14.46% in PCC, normalized root-mean-square error (NRMSE), and R^2 metrics, respectively.
3) In addition, our proposed method has exhibited favorable computational performance, laying the groundwork for preliminary research in prosthetics and facilitating natural interaction for users.

II. Methods

A. Dataset

To demonstrate the superiority of the SCTNet proposed in this article, we evaluated SCTNet on the Ninapro DB2 dataset. The Ninapro DB2 dataset [25] is a public open-source dataset that records 40 healthy subjects performing 49 different hand movements, each movement is repeated six times, each lasts about 5 s, and then rests for 3 s. During the execution of the motion, the Delsys Trigno wireless EMG sensor was used to collect the sEMG signals of 12 electrodes at a sampling frequency of 2 kHz, and the CyberGlove data glove was used to collect 22 finger joint angles at a sampling frequency of 20 Hz. The distribution of EMG sensors is: eight electrodes are equally spaced around the forearm at the height of the radio-humeral joint; two electrodes are placed on the main activity spots of the flexor digitorum superficialis and of the extensor digitorum superficialis; and two electrodes were also placed on the main activity spots of the biceps brachii and of the triceps brachii.

We meticulously chose 20 subjects from the Ninapro DB2 dataset to encompass a broad spectrum of physiological characteristics (13 males and seven females, age 23–45 years, 17 right-handed and three left-handed, ranging from 150 to 192 cm in height and 44 to 98 kg in weight). The specific physiological information of the subjects can be seen in Table I.

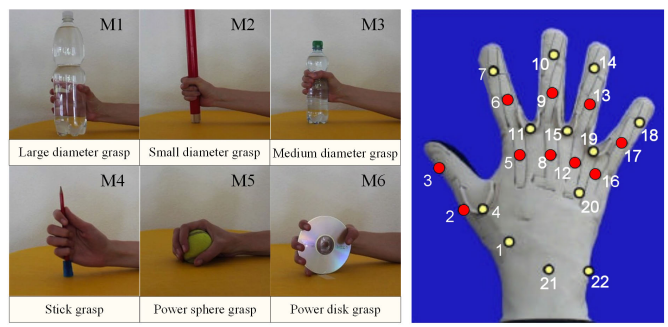
Because grasping movements are the most common hand movements in daily life, we selected six typical grasping movements [17] according to different shapes and diameters, as shown in Fig. 1(a). As shown in Fig. 1(b), we choose 10c proximal interphalangeal joints (PIP) and metacarpophalangeal joints (MCP) for estimation because they are the main moving joints in grasping movements.

B. Data Preprocessing

The finger joint angles were initially resampled to 2000 Hz to synchronize with the sEMG signal. Given the inevitability of noise, it is imperative to apply noise reduction techniques to the acquired sEMG signal [26]. We used a fourth-order Butterworth filter ranging from 10 to 400 Hz to eliminate dc components and high-frequency noise [27], [28], alongside a 50-Hz notch filter to counter power frequency interference [29].

TABLE I
Physiological Information of Subjects

Subjects	Gender	Dominant hand	Age	Heighht	Weight
01	Male	Right	29	187	75
02	Male	Right	29	183	75
04	Female	Left	30	154	50
05	Male	Right	25	175	70
06	Male	Right	35	172	79
08	Male	Right	45	173	73
09	Male	Right	23	172	63
11	Female	Right	32	150	54
14	Female	Right	30	173	59
15	Male	Right	30	169	58
19	Female	Right	31	158	52
21	Male	Right	32	170	75
22	Female	Left	28	162	54
23	Male	Right	25	170	66
26	Male	Left	30	186	90
27	Male	Right	29	170	65
30	Male	Right	30	173	68
31	Male	Right	29	185	98
33	Male	Right	31	192	78
35	Female	Right	24	170	52
36	Female	Right	27	155	44



(a) (b)

Fig. 1. (a) Six grasping movements. (b) CyberGlove data gloves, the red dots represent ten finger joints we want to predict.

Feature extraction from the sEMG signal is crucial for enhancing model accuracy and robustness. We segmented both the sEMG signal and joint angles into continuous sequences using a shifted window with a time window of 100 ms and an increment window of 0.5 ms [18]. Root-mean-square (rms) features were extracted from the sEMG signals of each sequence, with the last set of joint angles chosen for each sequence [30]. The calculation formula for rms is as follows:

$$\text{rms} = \sqrt{\frac{1}{N_t}\sum_{i=1}^{N_t} x_i^2} \tag{1}$$

where N_t is the length of the window, and x_i is the value of the sEMG signal at time i.

We normalize the rms features and joint angles of the sEMG signal to the range [0, 1] [31]. The sEMG features and joint angles are divided into sample sets using a sliding window with a time window of 100 ms and an increment window of 100 ms, where the last set of joint angles of each window is selected as labels for continuous motion estimation. The data dimension of the sample set is [1, 200, 12], and the label is [1, 10]. Among them, 1 represents the number of samples, 200 represents the sequence length, 12 represents the number of channels, and 10 represents the number of joints.

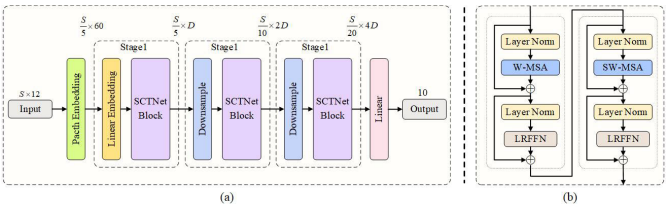


Fig. 2. (a) Architecture of SCTNet. (b) SCTNet Block. W-MSA and SW-MSA are window-based self-attention modules and shifted-window-based self-attention modules, respectively. LEFFN is a locally enhanced feedforward neural network module.

C. Shifted Windows and Convolution Network

An overview of the SCTNet architecture is presented in Fig. 2(a). Given the input sEMG feature matrix $I \in R^{(S/5)\times 5\times C}$, S is the sequence length and C is the number of channels. It first splits the input sEMG feature matrix into several tokens through patch embedding layer to realize the concatenation of adjacent feature values in the channel dimension. We use a 1-D convolution with a convolution kernel of 5 and a step size of 5 to achieve $I_p \in R^{(S/5)\times 5\times C}$. Then these tokens are projected to any dimension (denoted as D) through the linear embedding layer, $I_t \in R^{(S/5)\times D}$. These tokens calculate SW-MSA through the SCTNet Block layer, which can improve the representation ability of features. The output of "stage1" is $I_1 \in R^{(S/5)\times D}$. Afterward, downsampling is performed through the downsample layer to expand the receptive field range of each token, which is realized using a 1-D convolution with a convolution kernel of 2, a step size of 2, and an output channel of 2-D. The output of "stage2" is $I_2 \in R^{(S/10)\times 2D}$. And so on, the output of "stage3" is $I_3 \in R^{(S/20)\times 4D}$. Finally, a fully connected layer is used to complete the mapping from the feature space to the target value. In this study, C was set to 12 and D was set to 120.

1) SCTNet Block: The SCTNet Block as shown in Fig. 2(b) consists of two successive modules, each of which has two sublayers. The first module contains window-based self-attention (W-MSA) and LEFFN layer, and the second module contains SW-MSA and LEFFN layer. A layernorm (LN) layer is applied before each sublayer and a residual connection is applied after the sublayer. SCTNet Block is computed as

$$\widehat{x}^i = \mathrm{WMSA}\big(\mathrm{LN}\big(x^{i-1}\big)\big) + x^{i-1} \tag{2}$$

$$x^i = \mathrm{LEFFN}\big(\mathrm{LN}\big(\widehat{x}^i\big)\big) + \widehat{x}^i \tag{3}$$

$$\widehat{x}^{i+1} = \mathrm{SWMSA}\big(\mathrm{LN}\big(x^i\big)\big) + x^i \tag{4}$$

$$x^{i+1} = \mathrm{LEFFN}\big(\mathrm{LN}\big(\widehat{x}^{i+1}\big)\big) + \widehat{x}^{i+1} \tag{5}$$

where $\widehat{x}^i$ and x^i present the output features of the (S)W-MSA module and the LEFFN module for block i, respectively.

2) Multihead Attention: As shown in Fig. 3, the self-attention mechanism first learns three weight matrices W_Q, W_K, and W_V through the fully connected layer to map the input vectors into query vectors (Q), key vectors (K), and value vectors (V). The dot product of the query vector and the transpose of the key vector are then computed to obtain an attention score which measures the correlation between vectors. We scale the attention score by $(1/\sqrt{d_k})$ to prevent vanishing gradients. Afterward, the attention score is normalized to the interval [0, 1] through the softmax operation to obtain the attention distribution which measures the network's attention to the sequence vector. Finally, the value vector is weighted and summed using the attention distribution. The self-attention mechanism can model the global dependencies between any two vectors in sequence data to extract global features. The self-attention mechanism is computed as

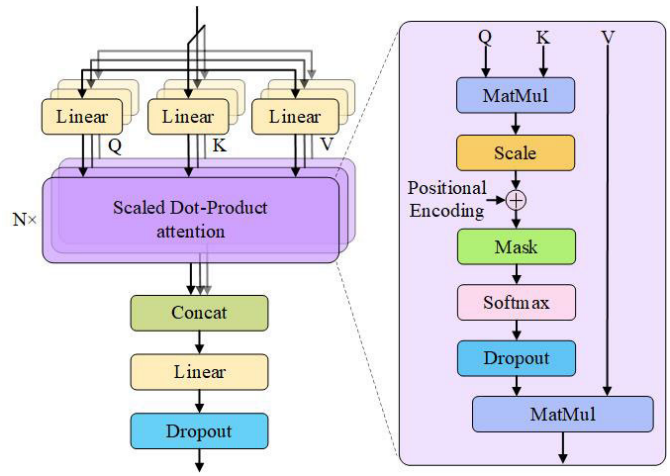


Fig. 3. Multihead attention mechanism.

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V. \tag{6}$$

We adopt a multihead attention mechanism to enhance the performance of the attention layer. We project the input vector into various subspaces using various linear layers, calculate self-attention for each, connect the outputs, and then project the output values again to get the final output value. Multihead attention mechanism is computed as

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)W^O \tag{7}$$

where

$$\mathrm{head}_i = \mathrm{Attention}\left(QW_i^Q, KW_i^K, VW_i^V\right) \tag{8}$$

where the weight matrix $W_i^Q \in R^{d_{\mathrm{model}}\times d_k}$, $W_i^K \in R^{d_{\mathrm{model}}\times d_k}$, $W_i^V \in R^{d_{\mathrm{model}}\times d_v}$, and $W^O \in R^{hd_v\times d_{\mathrm{model}}}$. In this work, h in the three SCTNet Blocks we use is 2, 4 and 8, and we use $d_k = d_v = d_{\mathrm{model}}/h$.

3) Shifted Window-Based Self-Attention: The author of Swin Transformer proposed shifted window self-attention to reduce computational complexity because the computational complexity of global self-attention is quadratically related to the sequence length. In this work, we follow this design idea. As shown in Fig. 4(a), the W-MSA module evenly divides the 1×8 feature map into two windows of size 1×4 ($M = 4$), and only calculates self-attention within the window. The SW-MSA module is used to first shift the window to the right by M/2 tokens, and then calculate the self-attention of the window to realize the information transmission between the windows. In this work, the window size was set to $M = 10$.

Due to the shifted window's inconsistent effect on the amount of tokens in the window, a cyclic shift as shown in Fig. 4(c) is necessary. After shifting, the window may consist of several subwindows that are not adjacent in the feature map, so a masking mechanism is used to limit the

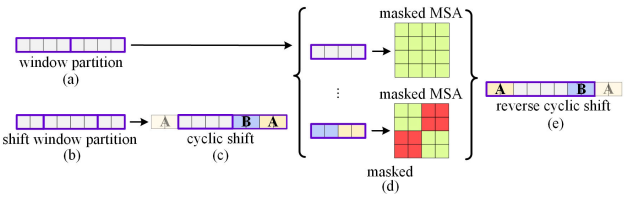


Fig. 4. Shifted window masking mechanism. (a) Window partition module. (b) Shift window partition module. (c) Cyclic shift module. (d) Masked MSA module.

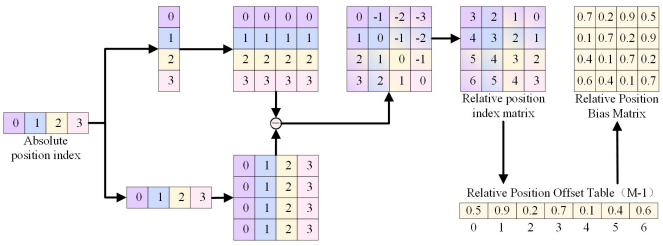


Fig. 5. Relative position encoding.

self-attention computation within each subwindow. As shown in Fig. 4(d), we set a large negative number in the red rectangle and superimpose it with the attention score. The attention distribution obtained by this part through the softmax function is 0 to shield the information of different regions. As shown in Fig. 4(e), reset the data after the calculation is completed.

4) Relative Position Encoding: Since sEMG signals are time-series data, the relative positional relationship between tokens needs to be considered when calculating window-based self-attention. We use relative positional encoding to introduce time-series information to the model. As shown in Fig. 5, assuming the window size is M, we first define the absolute position index of each token in the window, and then calculate the difference between each token and the absolute position index of all the tokens in the window to obtain the relative position matrix. We add M-1 to all the elements in the matrix to eliminate nonzero elements to obtain a relative position index matrix. We set up a learnable relative positional bias table to learn the best positional encoding from which to look up parameters using the relative positional index to obtain the relative positional bias matrix. Finally, the relative position bias matrix is added to the attention scores to learn the timing information of sEMG signals through relative position encoding. The final self-attention mechanism is computed as

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + B\right)V. \tag{9}$$

5) Locally Enhanced Feedforward Neural Network: As shown in Fig. 6, we propose an LEFFN layer. First, we map the feature vector to a high-dimensional space by 1×1 convolution. Since downsampling will lose part of the local information while expanding the receptive field, we use 1×3 depthwise convolution to extract the local features in the channel to compensate. Depthwise convolution is a special case of group convolution, where the number of groups is equal to the number of channels, and depthwise convolution can reduce the amount of calculation compared with ordinary convolution. We apply residual connections to stabilize the data and use the GELU activation function to improve the nonlinear representation ability of the model. Finally, the feature vector is mapped back to the low-dimensional space by 1×1 convolution. LEFFN can enhance the ability of the model to extract local features and combine the advantages of CNN to extract local features and transformer to extract global features. LEFFN is computed as

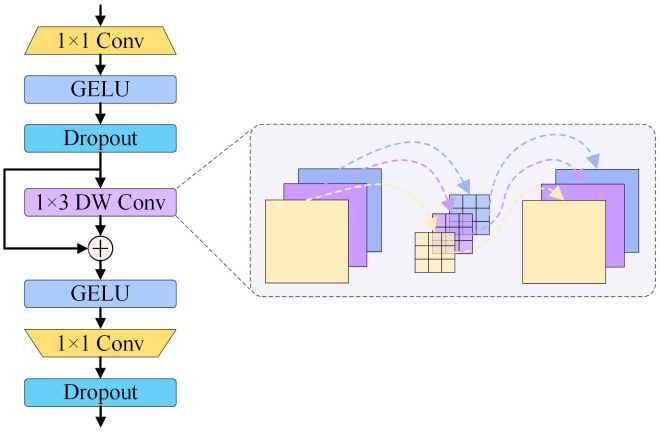


Fig. 6. Locally enhanced feedforward neural network.

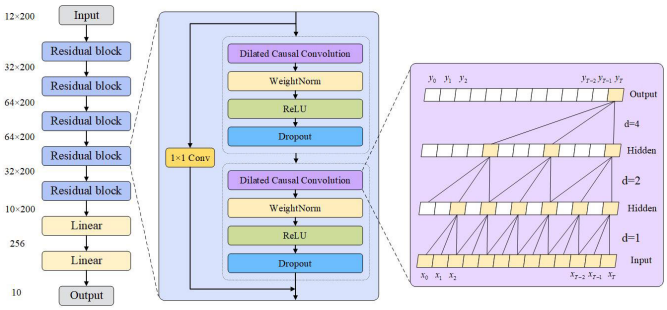


Fig. 7. Architecture of LS-TCN.

$$x' = \text{Dropout}(\text{GELU}(xW_1 + b_1)) \tag{10}$$

$$x'' = x' + \text{DWConv}(x') \tag{11}$$

$$\text{LEFFN}(x) = \text{Dropout}(\text{GELU}(x''W_2 + b_2)). \tag{12}$$

6) LS-TCN: For comparison, the LS-TCN was also included in this study. Chen et al. [20] used the TCN model proposed by Bai et al. [19] for continuous motion estimation on the Ninapro DB2 dataset. As shown in Fig. 7, the LS-TCN model is composed of causal dilated convolution and residual connection and has the ability to model long-term sequences. The LS-TCN model stacks five layers of TCN with a convolutional kernel size of 1×31, the convolutional channels are [32, 64, 64, 32, 10], and the dropout is set to 0.2 to prevent overfitting. Finally, ten joint angles are output through two fully connected layers.

7) CNN-Attention: For comparison, the CNN-Attention was included in this study. Geng et al. [24] proposed to use the CNN-Attention model for the Ninapro DB2 dataset for continuous motion estimation. As shown in Fig. 8, Geng used the multiscale convolution structure to extract both large-scale and small-scale features simultaneously and multihead attention structure to estimate the finger joint angles.

8) ConvLSTM: For comparison, the ConvLSTM was included in this study. Guo et al. [18] applied the ConvLSTM model proposed by Shi et al. [32] to the Ninapro DB2 dataset for continuous motion estimation. In the original article, 100 subframe rms features are used as the input of a time step.

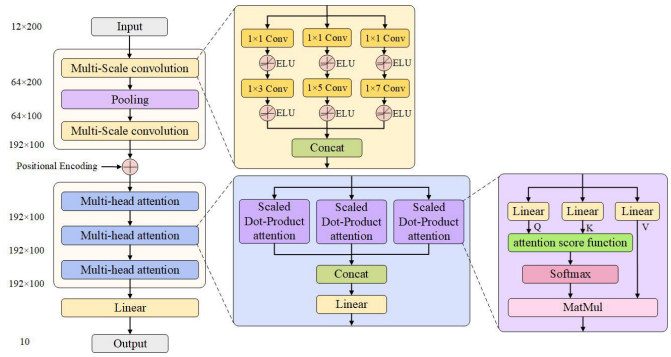


Fig. 8. Architecture of CNN-Attention.

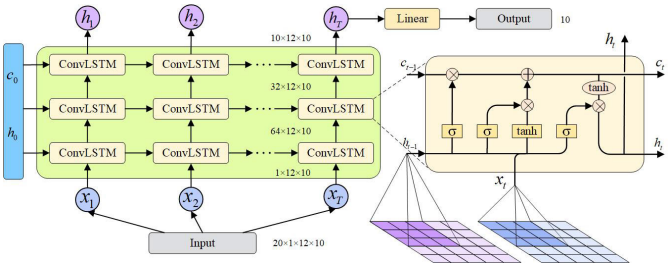


Fig. 9. Architecture of ConvLSTM.

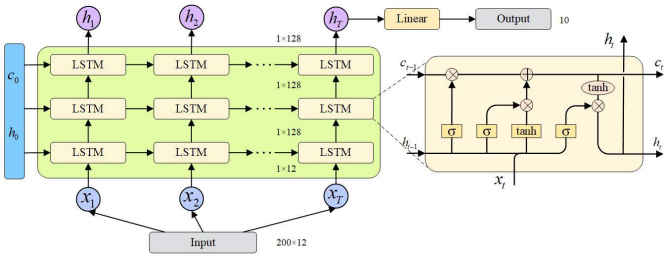


Fig. 10. Architecture of LSTM.

Since the ConvLSTM model has the ability to carry historical time information, the sEMG signal used for modeling will be much larger than 200 ms. To unify variables as shown in Fig. 9, this article splits the rms features of 100 subframes are divided into 20 rms features of five subframes as the input of the model. The rest of the structure is consistent with the original text, such as stacking three layers of ConvLSTM, and the hidden units of each layer are set to 64, 32, and 10 respectively. All the convolution operations use a 2-D invariant convolution with a convolution kernel of 3 and a step size of 1. Finally, ten joint angles are output through one fully connected layer.

9) LSTM: For comparison, the LSTM was also included in this study. Wang et al. [17] applied the LSTM model proposed by Hochreiter and Schmidhuber [33] to the Ninapro DB2 dataset for continuous motion estimation. The architecture of LSTM is shown in Fig. 10. Three LSTMs are stacked to increase the depth of the model. The hidden units of each layer are set to 256, and the dropout rate is set to 0.3 to prevent overfitting. Finally, ten joint angles are output through a fully connected layer.

D. Performance Metrics for Evaluation

The Pearson correlation coefficient (PCC) [34] can measure the linear relationship between two datasets. We use PCC to evaluate the correlation between the estimated joint angle and the real joint angle. The calculation formula is

$$\mathrm{PCC} = \frac{\sum_{i=1}^{N}\left(\theta_{\mathrm{est}} - \overline{\theta_{\mathrm{est}}}\right)\left(\theta_{\mathrm{real}} - \overline{\theta_{\mathrm{real}}}\right)}{\sqrt{\sum_{i=1}^{N}\left(\theta_{\mathrm{est}} - \overline{\theta_{\mathrm{est}}}\right)^2}\sqrt{\sum_{i=1}^{N}\left(\theta_{\mathrm{real}} - \overline{\theta_{\mathrm{real}}}\right)^2}} \times 100\% \tag{13}$$

where N, θ_{est}, $\overline{\theta_{\mathrm{est}}}$, θ_{real}, and $\overline{\theta_{\mathrm{real}}}$ present the number of samples, the value of estimated joint angle, the mean value of estimated joint angles, the value of real joint angle, and the mean value of real joint angles, respectively. The closer the PCC value is to 1, the stronger the correlation between the predicted joint angle and the actual joint angle, indicating higher estimation accuracy.

Root-mean-square error (RMSE) [35] cannot comprehensively measure the overall performance of the algorithm because the angular range of each joint is different. We use normalized RMSE to evaluate the deviation between the predicted joint angle and the actual joint angle to reflect the degree of discretization of the data. The calculation formula is

$$\mathrm{NRMSE} = \frac{1}{\theta_{\mathrm{MAX}} - \theta_{\mathrm{MIN}}}\sqrt{\frac{\sum_{i=1}^{N}\left(\theta_{\mathrm{eat}} - \theta_{\mathrm{real}}\right)^2}{N}} \times 100\% \tag{14}$$

where θ_{MAX} and θ_{MIN} denote the maximum and minimum values of the real joint angle, respectively. The smaller the NRMSE, the smaller the deviation between the estimated joint angle and the real joint angle, which means the higher the estimation accuracy.

Coefficient of determination (R^2) is a comprehensive evaluation index of the regression model, which is used to evaluate the fitting degree of the model. The larger the R^2 value, the better the fitting degree. The calculation formula is

$$R^2 = 1 - \frac{\sum_{i=1}^{N}\left(\theta_{\mathrm{real}} - \theta_{\mathrm{eat}}\right)^2}{\sum_{i=1}^{N}\left(\theta_{\mathrm{real}} - \overline{\theta_{\mathrm{real}}}\right)^2} \times 100\%. \tag{15}$$

III. Results

To conduct training and testing experiments, we partitioned the first four trials of each action into the training set and the final two trials into the testing set. The SCTNet was developed using the PyTorch framework, alongside LS-TCN, CNN-Attention, ConvLSTM, and LSTM models. These models were trained on the same GPU (Titan) and compared in continuous finger joint motion. Single motion estimation experiments were performed on a computer CPU with an Intel[1] Core[2] i7-1065G7 processor using PyCharm 2020 software. The optimization method used during training was Adam, with the learning rate set to 1e-4 and a batch size of 8.

The SCTNet's parameters were optimized, achieving peak performance with a dropout set to 0.5, while other models' dropout rates followed their original descriptions. To comprehensively evaluate model performance, we used PCC, normalized root mean square error (NRMSE), and R^2 as evaluation metrics in this study.

[1]Registered trademark.
[2]Trademarked.

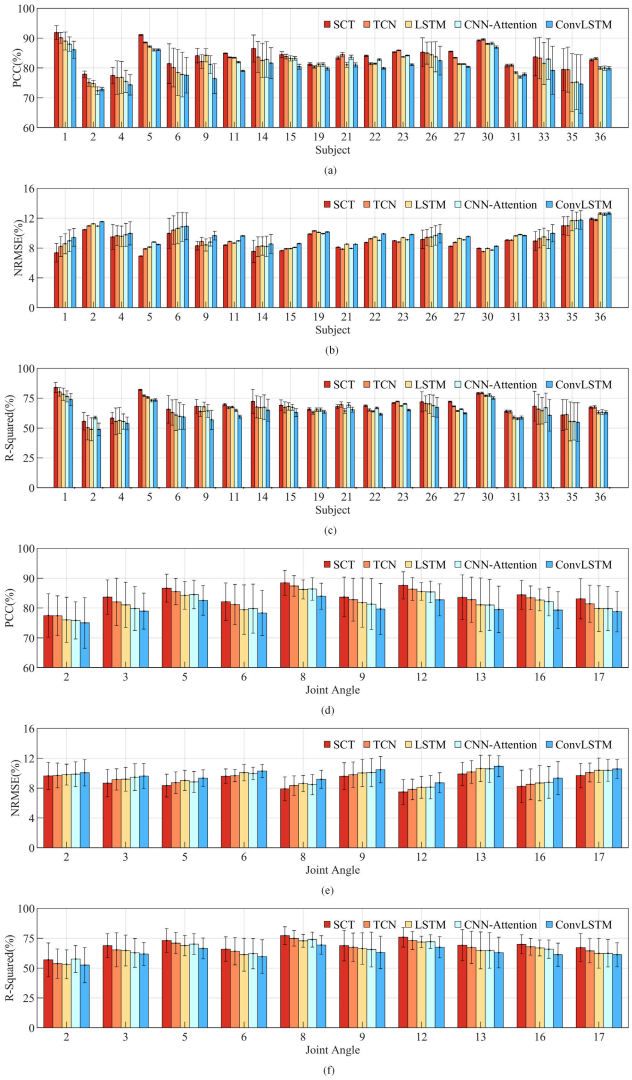


Fig. 11. Continuous motion estimation performance of different models for different subjects and different finger joints. (a) Results of PCC for different subjects. (b) Results of NRMSE for different subjects. (c) Results of R^2 for different subjects. (d) Results of PCC for different joints. (e) Results of NRMSE for different joints. (f) Results of R^2 for different joints.

The performance of five deep learning models for continuous motion estimation for various subjects and finger joints is displayed in Fig. 11. For 20 different subjects, the prediction accuracy of each subject is shown in Fig. 11(a)–(c). SCTNet performed better than the other four methods in PCC, NRMSE, and R^2 for 85% of the subjects. The PCC of SCTNet on all the subjects is greater than 77%, the NMRSE is less than 11%, and R^2 is greater than 57%. This shows that our method has strong generalization ability for different subjects. In addition, the prediction variance of SCTNet across all the subjects is consistently lower compared with the other four methods, underscoring the strong stability of our proposed method across diverse subjects.

For ten different joints, the prediction accuracy of each joint angle is shown in Fig. 11(d)–(f). SCTNet performed better than the other four methods in PCC, NRMSE, and R^2 for all the joints. The accurate prediction of ten joint angles across the dataset of 20 subjects underscores the robust generalization performance of our model. Through comprehensive statistical analysis across various subjects and finger joints, it becomes evident that our proposed SCTNet outperforms in terms of accuracy, generalization, and stability.

TABLE II
OVERALL PERFORMANCE OF DIFFERENT METHODS ON CONTINUOUS MOTION ESTIMATION OF FINGER JOINTS (MEAN RESULTS AND STANDARD DEVIATION)

Model	PCC(%)	NRMSE(%)	R2(%)	Times(ms)
SCTNet	**84.35±6.92**	**8.85±1.95**	**69.65±12.33**	9.795
LS-TCN	83.35±7.15	9.07±2.02	67.52±15.36	**9.364**
LSTM	81.97±7.22	9.41±2.08	66.07±11.70	40.01
CNN-Attention	82.09±7.52	9.33±2.17	66.64±11.98	47.22
ConvLSTM	80.46±7.24	9.73.02±2.11	63.53±12.83	49.18

Table II shows the overall performance of five deep learning models on continuous motion estimation of finger joints. It can be seen from the table that the mean and standard deviation of PCC, NRMSE, and R^2 of the SCTNet are 84.30% ± 6.92%, 8.85% ± 1.95%, and 69.65% ± 12.33%, respectively. Compared with other methods, the optimal results of the SCTNet have increased by 1% (PCC), 0.15% (NRMSE), and 2.13% (R^2) in each indicator. Statistical analysis proves that our proposed SCTNet outperforms other models in every evaluation index.

The prediction results of the ten finger joint angles of the SCTNet on subject 1 are shown in Fig. 12, the blue line represents the actual finger joint angle value, and the orange line represents the predicted value. Each subimage comprises six grasping actions, each executed twice. The significant angle variation between different grasping actions poses a challenge to the continuous motion estimation. Nevertheless, as depicted in the figure, the SCTNet's predicted values closely align with the actual finger joint angle curve, showcasing its effectiveness even under such challenging conditions.

It can be seen from Fig. 11(d)–(f) that the predicted effect of proximal interphalangeal joint angles (6, 9, 13, 17) is worse than that of metacarpal and phalangeal joint angles (5, 8, 12, 16). It is difficult for the model to extract features related to the angle of the proximal interphalangeal joint from sEMG, which may be the main reason for the decreased accuracy. It can be seen from Fig. 11(d) to (f) that due to the individual differences in the subjects, especially the predicted effect of the five deep learning models on subject 4 has decreased. We can roughly judge from the height and weight that subject 4 is petite and that the finger length is very likely to be relatively short. When she grasped the same object as other subjects, the change from the angle of the finger joints was small, which can also be confirmed from the original joint angle curve. Difficult identification of sEMG features adds to the difficulty of the prediction task. Nevertheless, the effect of our proposed SCTNet is significantly better than other models, the reason may be we adopt the transformer architecture to model the long-term dependencies of time series through the attention mechanism to capture global features, use layer-by-layer downsampling to improve the receptive field range of features, and learn the timing information of sEMG signals through learnable relative position encoding. We propose an

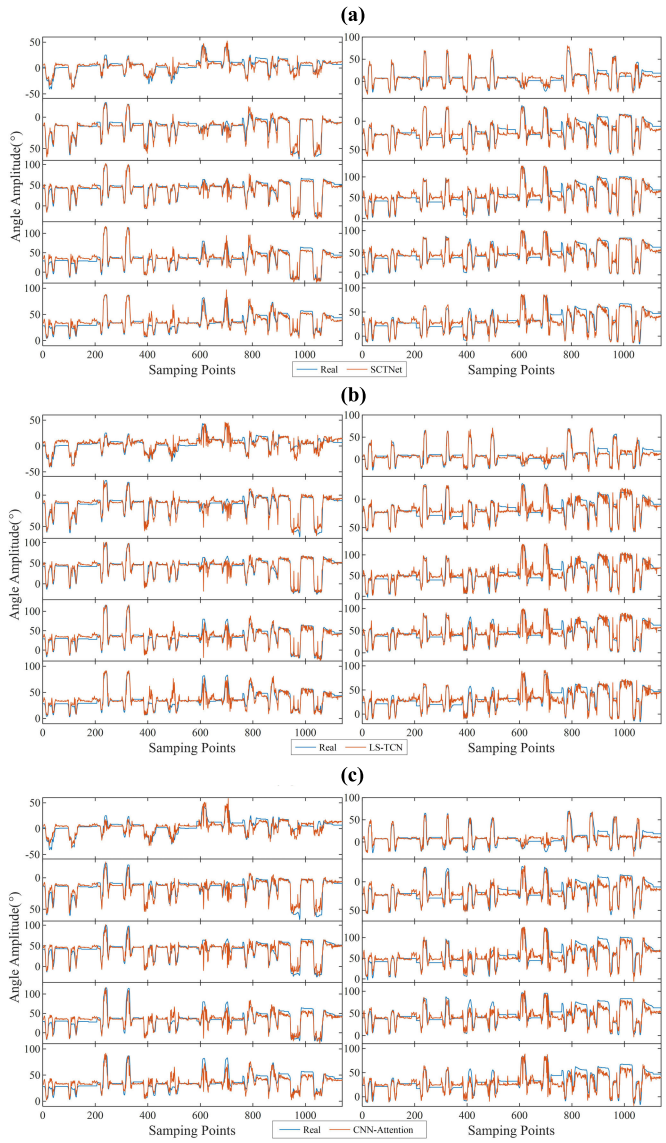


Fig. 12. Estimated joint angles against measured curves for subject 1, when using (a) SCTNet, (b) LS-TCN, and (c) CNN-Attention.

LEFFN layer, which combines the advantages of CNN for extracting local features and transformer for extracting global features and enhances the representation ability of features to improve the accuracy of motion estimation.

IV. Discussion

A. Optimization Experiment for the Shifted Window

Experimental validations were conducted on different shifted window lengths based on the SCTNet network in this article. Experiments were performed for shifted window lengths of 2, 6, 10, 14, and 18. The impact of these lengths was measured using four metrics: PCC, NRMSE, R^2, and real-time computational performance. The results are presented in Table III.

From the table, we observe that as the window length decreases, there is a slight increase in the metric data, but real-time computational performance decreases. Conversely, as the window length increases, there is a decrease in the metric data, but there is a slight improvement in real-time computational performance. To balance the accuracy of continuous angle estimation and real-time computational performance, we choose a shifted window length of 10.

TABLE III
Optimization Experiment for the Shifted Window

Length	PCC(%)	NRMSE(%)	R2(%)	Times(s)
2	84.52±6.76	8.79±1.97	69.12±12.81	9.915
6	84.46±6.54	8.82±1.98	69.87±12.38	9.835
10	84.35±6.92	8.85±1.95	69.65±12.33	9.795
14	84.11±7.22	9.03±1.96	68.13±11.27	9.745
18	83.94±7.52	9.45±1.86	68.13±12.13	9.725

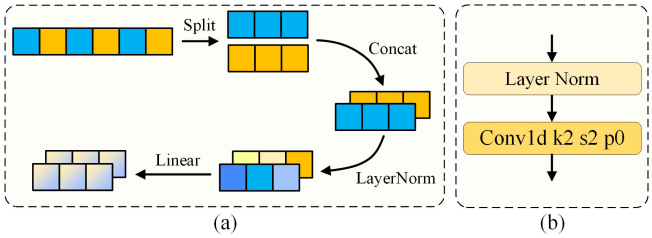


Fig. 13. Downsampling structure (a) PM and (b) 1 × 2 convolution with stride 2.

TABLE IV
Ablation Experimental Results for Position Encoding and Downsampling Structures

Model	PCC(%)	NRMSE(%)	R2(%)
Transformer	82.28±6.32	9.46±1.76	65.64±11.05
Swintf-APC-PM	82.85±5.82	9.28±1.61	66.94±10.34
Swintf-APC-CONV	82.27±6.23	9.36±1.63	66.34±10.67
Swintf-RPC-PM	**83.43±6.16**	9.21±1.70	66.89±12.37
Swintf-RPC-CONV	83.40±6.36	**8.85±1.95**	**69.65±11.81**

B. Ablation Experiment of Positional Encoding and Downsampling Structure

We have followed the design idea of the Swin Transformer model to reconstruct each module and have obtained the SwinTf model. To achieve a model with better effectiveness, we have obtained the SCTNet by improving some structures. To illustrate the effectiveness of each part of the model, this article has designed an ablation experiment based on the SwinTf model. Ablation experiments mainly consist of positional encoding, downsampling, and LEFFN. We have first explored the position encoding and downsampling structure. The position encoding method adopts absolute position encoding (APC) and relative position encoding (RPC) respectively. The downsampling structure is shown in Fig. 13. Fig. 13(a) is our reconstructed patchmerging (PM) structure based on the Swin Transformer model, and Fig. 13(b) is implemented using a 1 × 2 convolution (CONV) with a stride of 2. In addition, we have experimented with a three-layer transformer encoder for comparison. The results of the five ablation experiments are shown in Table IV; we have weighted various evaluation indicators and think that the combination of relative position encoding and convolutional downsampling is better. We call this model the STNet.

C. Ablation Experiment of LEFFN Layer

Next, an ablation experiment was designed for LEFFN. As shown in Fig. 11, we try to introduce depthwise convolution and residual connections in the feedforward neural

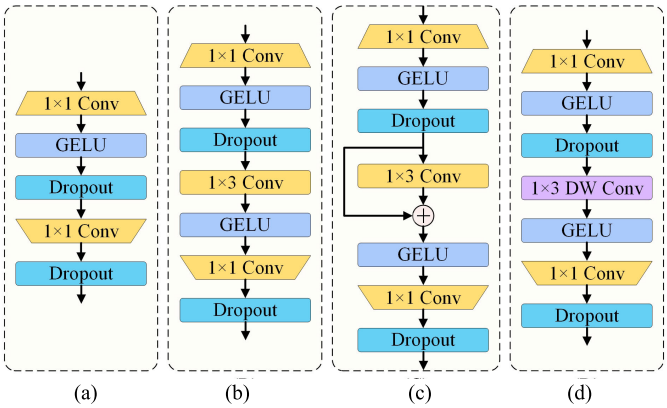


Fig. 14. Localized enhanced feedforward neural network layer ablation experiment. (a) FFN. (b) Regular convolution. (c) Regular convolution with residual connection. (d) Depthwise separable convolution.

TABLE V
ABLATION EXPERIMENTAL RESULTS FOR LEFFN

Model	PCC(%)	NRMSE(%)	R2(%)
STNet-A	83.40±6.36	9.04±1.73	68.12±11.81
STNet-B	83.31±6.13	9.12±1.75	67.87±11.38
STNet-C	83.33±6.37	9.10±1.71	67.88±11.57
STNet-D	83.11±6.22	9.23±1.76	67.03±11.27
SCTNet	**84.35±6.92**	**8.85±1.95**	**69.65±12.33**

network (FFN) layer of the SwinTf model to enhance the representation ability of features. In the ablation experiment, we additionally designed four structures as shown in Fig. 14 and demonstrated the effectiveness of this part through experiments with ordinary convolution and with or without residual connections. The results of five ablation experiments are shown in Table V, which demonstrates that the SCTNet which combines depthwise convolution and residual connection is more effective.

D. Real-Time Computing Performance

The real-time performance of the model in continuous motion estimation is worth noting, because the computational time consumption of deep learning models will increase due to more complex network structures. It can be seen from Table II that the average time consumption of a single motion estimation of the SCTNet is 9.795 ms, which is competitive in real-time compared with other methods. Coupled with the data preprocessing time of about 2.481 ms, it is much lower than the real-time requirement of myocontrol (200 ms) [36].

V. CONCLUSION

In this article, we propose a network of deep learning named SCTNet for continuous angle estimation of finger joints. The SCTNet uses the attention mechanism to model the long-term dependencies of the time series to capture the global features, uses the shifted window to constrain the computational complexity of the model, and uses the LEFFN layer to combine the local features extracted by CNN and the global features extracted by transformer to enhance the representation ability of features. We verified it on the Ninapro DB2 dataset and compared with LS-TCN, LSTM, CNN-Attention, and ConvLSTM. The experimental results show that SCTNet performs better than other models in PCC, NRMSE, and R^2. SCTNet also achieves competitive results in real-time computing performance. The SCTNet can describe joint motion with high precision and fine-granularity, which is very helpful for smooth, natural, and low-cost natural prosthetic trajectory estimation.

However, this work also has some limitations. First, the hand movements in this study are limited to six typical grasping movements, which are not enough to represent complex and changeable hand movements. In future work, we consider covering more hand movements when training the model, or using a variety of typical hand movements to train a general network to predict more complex hand movements through few-shot learning. Second, due to individual differences among subjects, the trained model is not very effective in continuous motion estimation across subjects. In the future, we consider using domain-adapted transfer learning methods to extract person-invariant features in sEMG signals to improve the recognition performance of the model across subjects. Finally, wrong electrode positioning can impact the proposed method. So how to correct electrode positioning is worth studying.

REFERENCES

[1] K. Reed and P. Kalu, "An assessment of bamboo as a potential low-cost material for exoskeleton design in normal walking," *Int. J. Phys. Therapy Rehabil.*, vol. 2, no. 1, p. 111, Feb. 2016, doi: 10.15344/2455-7498/2016/111.

[2] C. Li, G. Li, G. Jiang, D. Chen, and H. Liu, "Surface EMG data aggregation processing for intelligent prosthetic action recognition," *Neural Comput. Appl.*, vol. 32, no. 22, pp. 16795–16806, Nov. 2020, doi: 10.1007/S00521-018-3909-Z.

[3] T. Bao, S. Q. Xie, P. Yang, P. Zhou, and Z.-Q. Zhang, "Toward robust, adaptiveand reliable upper-limb motion estimation using machine learning and deep learning—A survey in myoelectric control," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 8, pp. 3822–3835, Aug. 2022, doi: 10.1109/JBHI.2022.3159792.

[4] T. M. Bittibssi, A. H. Zekry, M. A. Genedy, and S. A. Maged, "SEMG pattern recognition based on recurrent neural network," *Biomed. Signal Process. Control*, vol. 70, Sep. 2021, Art. no. 103048, doi: 10.1016/J.BSPC.2021.103048.

[5] S. Morita, K. Shibata, X. Z. Zheng, and K. Ito, "Prosthetic hand control based on torque estimation from EMG signals," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Aug. 2000, pp. 389–394, doi: 10.1109/IROS.2000.894636.

[6] R. Raj, R. Ramakrishna, and K. S. Sivanandan, "A real time surface electromyography signal driven prosthetic hand model using PID controlled DC motor," *Biomed. Eng. Lett.*, vol. 6, no. 4, pp. 276–286, Nov. 2016.

[7] Q. Ding, A. Xiong, X. Zhao, and J. Han, "A review on researches and applications of sEMG-based motion intent recognition methods," *Acta Autom. Sinica*, vol. 42, no. 1, pp. 13–25, 2016.

[8] G. Pomboza-Junez and J. H. Terriza, "Hand gesture recognition based on sEMG signals using support vector machines," in *Proc. IEEE 6th Int. Conf. Consum. Electron.*, Sep. 2016, pp. 174–178.

[9] K.-H. Park and S.-W. Lee, "Movement intention decoding based on deep learning for multiuser myoelectric interfaces," in *Proc. 4th Int. Winter Conf. Brain-Comput. Interface*, 2016, pp. 1–2.

[10] E. Rahimian, S. Zabihi, A. Asif, D. Farina, S. F. Atashzar, and A. Mohammadi, "TEMGNet: Deep transformer-based decoding of upperlimb sEMG for hand gestures recognition," 2021, *arXiv:2109.12379*.

[11] X. Lv et al., "Gesture recognition based on sEMG using multi-attention mechanism for remote control," *Neural Comput. Appl.*, vol. 35, no. 19, pp. 13839–13849, Jul. 2023.

[12] K. Li, J. Zhang, L. Wang, M. Zhang, J. Li, and S. Bao, "A review of the key technologies for sEMG-based human–robot interaction systems," *Biomed. Signal Process. Control*, vol. 62, Sep. 2020, Art. no. 102074.

[13] X. Xi et al., "Simultaneous and continuous estimation of joint angles based on surface electromyography state-space model," *IEEE Sensors J.*, vol. 21, no. 6, pp. 8089–8099, Mar. 2021.

[14] T. S. Buchanan, D. G. Lloyd, K. Manal, and T. F. Besier, "Neuromusculoskeletal modeling: Estimation of muscle forces and joint moments and movements from measurements of neural command," *J. Appl. Biomechanics*, vol. 20, no. 4, pp. 367–395, Nov. 2004.

[15] R. Alazrai, D. Alabed, N. Alnuman, A. Khalifeh, and Y. Mowafi, "SEMG-based approach for estimating wrist and fingers joint angles using discrete wavelet transform," in *Proc. IEEE EMBS Conf. Biomed. Eng. Sci.*, Dec. 2016, pp. 596–599, doi: 10.1109/IECBES.2016.7843519.

[16] C. Ma, W. Guo, L. Xu, and G. Li, "Finger joint angle estimation based on sEMG signals and deep learning method," in *Proc. IEEE Int. Conf. Real-Time Comput. Robot. (RCAR)*, Jul. 2021, pp. 638–643, doi: 10.1109/RCAR52367.2021.9517643.

[17] C. Wang, W. Guo, H. Zhang, L. Guo, C. Huang, and C. Lin, "SEMG-based continuous estimation of grasp movements by long-short term memory network," *Biomed. Signal Process. Control*, vol. 59, May 2020, Art. no. 101774.

[18] W. Guo et al., "Long exposure convolutional memory network for accurate estimation of finger kinematics from surface electromyographic signals," *J. Neural Eng.*, vol. 18, no. 2, Apr. 2021, Art. no. 026027.

[19] S. Bai, J. Zico Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," 2018, *arXiv:1803.01271*.

[20] C. Chen, W. Guo, C. Ma, Y. Yang, Z. Wang, and C. Lin, "SEMG-based continuous estimation of finger kinematics via large-scale temporal convolutional network," *Appl. Sci.*, vol. 11, no. 10, p. 4678, May 2021.

[21] A. Vaswani et al., "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 1–18.

[22] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[23] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[24] Y. Geng et al., "A CNN-attention network for continuous estimation of finger kinematics from surface electromyography," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 6297–6304, Jul. 2022.

[25] M. Atzori et al., "Electromyography data for non-invasive naturally-controlled robotic hand prostheses," *Sci. Data*, vol. 1, no. 1, pp. 1–13, Dec. 2014.

[26] W. Li, P. Shi, and H. Yu, "Gesture recognition using surface electromyography and deep learning for prostheses hand: State-of-the-art, challenges, and future," *Frontiers Neurosci.*, vol. 15, Apr. 2021, Art. no. 621885.

[27] S. Song, L. Yang, M. Wu, Y. Liu, and H. Yu, "Dynamic hand gesture recognition via electromyographic signal based on convolutional neural network," in *Proc. IEEE Int. Conf. Syst. Man, Cybern. (SMC)*, Oct. 2021, pp. 876–881.

[28] C. Ma et al., "A novel and efficient feature extraction method for deep learning based continuous estimation," *IEEE Robot. Autom. Lett.*, vol. 6, no. 4, pp. 7341–7348, Oct. 2021.

[29] D. Bu, S. Guo, and H. Li, "SEMG-based motion recognition of upper limb rehabilitation using the improved YOLO-v4 algorithm," *Life*, vol. 12, no. 1, p. 64, Jan. 2022.

[30] X. Ma, Y. Liu, Q. Song, and C. Wang, "Continuous estimation of knee joint angle based on surface electromyography using a long short-term memory neural network and time-advanced feature," *Sensors*, vol. 20, no. 17, p. 4966, Sep. 2020.

[31] A. R. Zangene, A. Abbasi, and K. Nazarpour, "Estimation of lower limb kinematics during squat task in different loading using sEMG activity and deep recurrent neural networks," *Sensors*, vol. 21, no. 23, p. 7773, Nov. 2021.

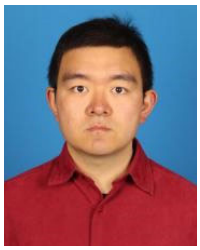
[32] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst.*, 2015, pp. 1–17.

[33] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.

[34] Y. Lu, H. Wang, B. Zhou, C. Wei, and S. Xu, "Continuous and simultaneous estimation of lower limb multi-joint angles from sEMG signals based on stacked convolutional and LSTM models," *Expert Syst. Appl.*, vol. 203, Oct. 2022, Art. no. 117340.

[35] X. Wang et al., "SEMG-based consecutive estimation of human lower limb movement by using multi-branch neural network," *Biomed. Signal Process. Control*, vol. 68, Jul. 2021, Art. no. 102781.

[36] K. Englehart and B. Hudgins, "A robust, real-time control scheme for multifunction myoelectric control," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 7, pp. 848–854, Jul. 2003.



Zida An was born in Shenyang, Liaoning, China, in 1998. He received the B.S. and M.S. degrees in control theory and control engineering from Northeastern University, Shenyang, China, in 2021, where is currently pursuing the Ph.D. degree with the Faculty of Robot Science and Engineering.

His research interests include deep learning, intelligent control, and human–computer interaction.

Fei Wang (Member, IEEE) received the B.S. and M.S. degrees from Harbin Institute of Technology, Harbin, China, in 2000, and the Ph.D. degree in electrical engineering from the University of Tokushima, Tokushima, Japan, in 2004.

He has long been committed to researching and solving key scientific theoretical and technical problems in the field of intelligent robots such as mechatronics, multimodal perception and cognition, and human–machine interaction.

Chuanjian Ren received the master's degree from Northeastern University, Shenyang, China, in 2023.

His research focuses on key scientific theories and technical challenges in the field of intelligent robotics, including physiological signal perception and human–machine collaborative control.

Kun Zhang was born in Wuwei, Gansu, China, in 1998. He received the B.S. degree in automation from Northeastern University, Shenyang, China, in 2021, where he is currently pursuing the M.S. degree with the Faculty of Robot Science and Engineering.

His research interests include deep learning, rehabilitation robot, intelligent control, and human–computer interaction.

Yao Pei was born in Yangquan, Shanxi, China, in 2001. She is currently pursuing the master's degree with the School of Information Science and Engineering, Northeastern University, Shenyang, China.

Her research interests include deep learning, intelligent control, and human–computer interaction

Qichuan Ding (Member, IEEE) received the B.E. degree in information and computing science from Shandong University, Jinan, China, in 2007, and the Ph.D. degree in pattern recognition and intelligent system from Shenyang Institute of Automation, Chinese Academy of Sciences, Shenyang, China, in 2015.

He is currently an Associate Professor with the Faculty of Robot Science and Engineering, Northeastern University, Shenyang. His main research interests include human–robot interaction, human action recognition, and rehabilitation robots.

Shuai Han received the M.M. degree from China Medical University, Shenyang, China, in 2015, where he is currently pursuing the Ph.D. degree. In 2018, he was the first doctor to use a Chinese-made robot to perform neurosurgery in northeast China. Nearly 1000 neurosurgical robotic operations have been completed. He pioneered the use of dental registration to improve robotic surgery accuracy and reduce trauma. He has published ten articles as the first or corresponding author. His research interests include robotic minimally invasive neurosurgery and brain–computer interface (BCI).